

- Almost all kinds of crops can be grown on *red soil*, although it seems to be more suitable for the cultivation of rice, ragi, tobacco and vegetables.
- Laterite soils are suitable, among others, for rice and sugarcane.

Different types of soils distributed evenly throughout the country, abounding in fertility and higher yields, and highly responsive to improved inputs are found in the country. For example, we find the desert-like region of Rajasthan on the one hand and the rich cultivable land of Gujarat on the other. The variety of soils coupled with the fact that we have in the East the world's highest rainfall zone and in the West one of the driest regions along with every shade of climate throughout the country, makes possible the production of almost every kind of crop starting from those of the temperate zone to tropical production.

However, through constant use the quality of these soils has deteriorated. Moreover, large tracts of land have been eroded. It has been estimated that about 106 million hectares is suffering from varying degrees of soil degradation. Localised soil waterlogging and salinity are most severe in India (27 per cent of irrigated land), Pakistan (20 per cent) and China (15 per cent).

Although our Plans have given priority to soil conservation and land stock improvement, we can identify the following difficulties in containing the degradation of land resources and bringing them back to productive uses:

- i) issues related to management of community land;
- ii) lack of infrastructural development;
- iii) high investment and long gestation;
- iv) non-availability of institutional finance due to low credit worthiness of the beneficiaries having marginal and sub-marginal lands.

4.3 CROPPING PATTERN IN INDIA

By the term 'crop pattern' we mean the crop - wise distribution of the cultivated land area between different crops, the changes in the distribution over time, and the determinants of this distribution. The crop pattern in India presents some distinct features as follows:

4.3.1 Crop Diversity

Being a country almost of a continental size, India is endowed with a variety of soils, which make it possible to grow almost every kind of crop. The various crops being cultivated in India can be grouped into (a) food crops, (b) fibres, (c) oilseeds, (d) drugs and spices. Food crops are both of a cereal and non-cereal type. Among the cereals the more important are rice, wheat, besides maize, jowar, bajra, barley, etc. The more important non-cereal food crops are pulses like gram, moong, urad, etc. Among fibres, cotton and jute are the important crops. Among oilseeds, groundnut is the most important besides rapeseed and mustard. Similarly, a number of spices and medicinal crops are also cultivated throughout the country. Newer crops are emerging – products of tree crops, of the jungle and so on – which rakes in large sums of money while the going is good.

The Indian farmer has adopted a necessity-oriented approach. He has concentrated on food crops in preference to commercial crops: about three-fourths of the total

sown area in the country has been devoted to the production of food crops. Some of this has been induced by agricultural pricing policies of the government.

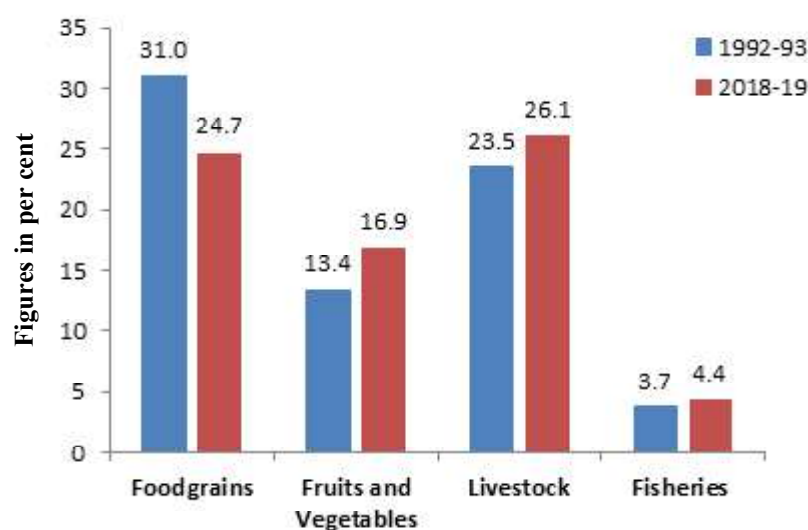


Fig. 4.1: Share in Value of Agricultural Output

Recent trends indicate that agriculture mix has been undergoing a silent and steady shift away from grains (refer Figure 4.1). With rising incomes, this change is driven by consumers' tastes and preferences. The composition of the production basket is also changing as is evident from the rising share of high value commodities (fruits, vegetables, fisheries, and livestock products). The value of output of these commodities has gone up from 40.6 per cent in Triennium Ending (TE) 1992-93 to 47.4 per cent in TE 2018-19, while the share of foodgrains has come down from 31 per cent to 24.7 per cent over the same period. This will be good for the farmers as it helps augment their incomes and achieve household food security.

Preponderance of Cereals among Food crops

Within the broad group of food crops, cereals like Wheat and Rice dominate. About 82 per cent of the land area under food crops has been put to the cultivation of cereals. The attraction of cereal crops, as already stated, is explained by their relatively better prices and less risk in production, and finally, the availability of better-quality seeds. As a result of this tendency per capita availability of cereals has improved from 334.2 gms. per day in 1951 to 407.4 gms. in 2018, and India has successfully transformed itself into a 'net exporter' of cereals from the position of a 'net importer', the per capita availability of pulses (otherwise a nutrition rich food) has declined from 60.7 gms. per day in 1951 and 74.9 gms in 1959 to only 35.5 gms in 2018. Since 2015 there has been a major break-through in production of pulses. Their total output has gone up by more than 50 per cent in this short-period.

Inferior cereals like jowar, bajra, maize, millets, barley, etc. account for a little less than one-fourth of the total area under cereal cultivation.

These crops are cultivated under dry conditions. Although the productivity is low, the crops are safer and are not sensitive to the vagaries of monsoon. Therefore, the farmers often have resorted to their cultivation as an insurance against misery arising out of crop failure.

4.3.2 Future Cropping Pattern in India

In India, the policy objectives regarding the future crop pattern should be based on the following considerations:

A) Rise in Income Levels

As the economy grows, the level of income will rise. It is only to be expected that this would involve an increasing demand for superior cereals. Simultaneously, as the size of India's middle class increases, there will be changes in consumption patterns leading to significantly increased demand for higher value crops. Therefore, in the long run it is imperative that crop pattern is adjusted to the changing requirements by shifting land under cultivation from inferior crops to superior crops.

B) Need for Diversification

There is a need for diversifying the crop pattern by encouraging the cultivation of different crops, due to the following reasons:

- It will ensure stability in over-all agricultural production, in as much as some crops may always compensate for fall in output of some other crops.
- The shift to high-valued crops will raise the total value of agricultural output.
- It will provide a base for increasing surplus of those products that are in increasing demand abroad. It will help to raise foreign exchange for the economy.
- The farmers with a reasonably-sized land, co-operatives with land of very small farmers pooled in, will gain from stable or higher incomes.

C) Need to adapt to Water Scarcity

It is high time that crop pattern changes should be introduced in regions increasingly faced with water shortages. Crops like rice and sugarcane have been described as "*parasites on water*". These crops may be substituted by light irrigated crops like millets, pulses, oilseeds, cotton, fruit, flowers and vegetables, especially in water scarce areas. There is no case in the present context of India's food economy to grow rice in Punjab and Haryana or Western Uttar Pradesh or sugarcane in Parts of Maharashtra.

D) Need to adapt to National Priorities

A "desirable cropping pattern" should contribute towards national priorities, namely: employment generation, sustainability, higher productivity ensuring stable farm income and food-security, combating climate change, etc. By implication, a desirable cropping pattern should be one which favours crops which are labour-intensive and have greater second-round employment effects. Simultaneously, given water as a binding constraint, the cropping pattern should attempt to maximise returns to this input and favour crops that are ecologically sustainable. The agriculture productivity should also be enhanced to make sure farmers are assured of minimum returns and that there are no food shortages in the times of climate extremities.

This last point calls for a little further attention. At a global level, India is one of the richest nations in terms of biological diversity. India supports 15,000 species